

The Corrections of Sub-picosecond Propagation Time Delays and the Allan Variance of Discontinuous Measurement for the Inter-satellite Time Transfer[†] ^{*}

He Ke-liang He Yu-ling[△] Wang Guo-yong

China Academy of Space Technology (Xi'an), Xi'an 710100

Abstract The asymmetric signal propagation paths between two satellites lead to the unequal time delays at receiving signal time, which should be accurately corrected when solving the clock difference of inter-satellite time transfer. The observation method of clock difference is analysed, the asymmetric signal propagation time delay caused by satellite motion is then derived and reduced, the delay caused by relativistic deflection of light is also calculated using the time transfer function method, and the residual error stability of the propagation time delay correction is evaluated, which improve the correction accuracy of asymmetric signal propagation time delay to 0.1 ps under the current condition of orbit determination. The autocorrelation function representation of Allan variance is extended, and a method of computing the Allan variance error with missing measuring data is proposed based on the statistical method. The effect of missing measuring data on frequency stability calculation of different noises is also analysed. Overall, a systematic correction approach of the propagation time delay and Allan variance with discontinuous measurement for the inter-satellite two-way time transfer has been constructed, which can be used in the clock difference correction of inter-satellite time transfer in the order of ps.

Key words time: time transfer—propagation delay—reference systems: atomic time—methods: statistical

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[△] heyuling.cast@163.com

1. INTRODUCTION

The high-precision inter-satellite and satellite-ground links of time and frequency transfer with long term stability are of great significance for improving the accuracy of time keeping, orbit determination, navigation, and basic physics measurement. Inter-satellite time transfer solves the inter-satellite clock difference (proper time difference) corresponding to a coordinate time on the basis that the terminals on the satellite emit and receive signals in accordance with a specific time sequence, measure two-way pseudo-ranges, local or inter-satellite time intervals between reception and emission, and correct the hardware and propagation delays. The asymmetric signal propagation paths between two satellites cause unequal delays when received, which causes the asymmetric propagation time delay. The uncorrected asymmetric propagation time delay will lead to additional measurement errors of pseudo-ranges and time intervals between reception and emission. The asymmetric signal propagation paths between two satellites are mainly due to relative motion of satellites, and also affected by relativistic deflection of light and non-simultaneous emission. The correction error of the asymmetric propagation time delay is already one of the main errors for the high-precision time transfer between satellites that are far apart. The asymmetric propagation time delay needs to be corrected more accurately so as to meet the requirement of improving further the accuracy of inter-satellite time and frequency transfer. The relative frequency stability of inter-satellite and satellite-ground time transfer can reach the level of 10^{-15} – $10^{-16}/\text{d}$, and the long term stability deteriorate throughout the links of the inter-satellite and satellite-ground time transfer in satellite navigation systems^[1–6], the two-way satellite time and frequency transfer (TWSTFT) between intercontinental stations^[7, 8], and the satellite-ground two-way time transfer^[9]. Correspondingly, the time measurement error is on the order of 0.1 ns in the current satellite-ground and inter-satellite time and frequency transfer links, and the time delay caused by the relative motion of satellites is considered briefly when correcting the asymmetric propagation time delay^[10–13]. It is the frontier of current research to improve the relative frequency stability of the inter-satellite and satellite-ground time and frequency transfer to the level of 10^{-17} – $10^{-18}/\text{d}$, of which the asymmetric propagation time delay needs to be corrected to the level of 0.1 ps even higher precision. The Atomic Clock Ensemble in Space (ACES) developed by European Space Agency will implement the satellite-ground microwave link of time and frequency transfer with a time stability at the level of 0.4 ps/300 s and 8 ps/d^[14–18]. The current correction methods for the asymmetric propagation delay are difficult to reach the level of 0.1 ps and higher precision.

The missing data with a period or not may be led by Earth occlusion or equipment failure for inter-satellite time transfer, which affects the clock difference measurement, performance evaluation, and clock difference prediction of atomic clocks on the satellite. The stability (Allan variance) calculation is almost unaffected when the averaging time τ is much less than the measurement period T in the case of a measurement sequence with periodic

missing data^[19]. Barnes et al.^[20, 21] and Wang et al.^[22] adopted the coefficient $B_2(T, \tau)$ to correct the stability calculation result when the averaging time τ is on the same order with the measurement period T and τ is less than T in the case of a measurement sequence with periodic missing data. For the case of τ larger than T or a measurement sequence with irregular missing data, Allan variance is difficult to calculate when the data missing proportion is high. Because the data period of inter-satellite time and frequency transfer is on the order of 10000 s, the data missing affects the stability calculation at averaging time one day, and there is a lack of relevant research on the correction methods.

A systematic approach to correct the propagation time delays and the Allan variance of discontinuous measurement for the inter-satellite time transfer is proposed in this paper. Firstly, the measurement method of clock difference is analyzed. Secondly, the asymmetric propagation delay caused by the relative motion of satellites is derived and reduced, and the delay caused by relativistic deflection of light is also calculated by using the time transfer function method, and the stability of residual error of the propagation time delay correction is evaluated. Finally, the autocorrelation function representation of Allan variance is extended, and a method of computing the Allan variance error with data missing is proposed based on the statistical method. The effect of data missing on frequency stability calculation of different noises is also analyzed.

2. A CORRECTION METHOD FOR THE ASYMMETRIC PROPAGATION DELAY

2.1 A Model for the Clock Difference of Inter-satellite Time Transfer

The Einstein's theory of general relativity predicts that the rates of ideal clocks at some positions with different gravitational potentials are different, and the synchronization of clocks at different positions needs to correct the clock rates according to a specified reference system. For example, the second definition of International Atomic Time Scale (TAI) chooses geoid as a reference, and the clock rates of the atomic clocks on navigation satellites are adjusted to be consistent with the proper time of ground atomic clocks. The non-rotating geocentric reference system is chosen to establish the observation equation of clock difference for the inter-satellite time transfer, the metric tensors of the geocentric reference system in the International Astronomical Union (IAU) 2000 resolutions are adopted^[23], and the time coordinate choose the Terrestrial Time (TT) which is defined on the geoid, namely the space-time coordinates are (cT, X_i) , where c , T , and X_i represent respectively the speed of light, the Terrestrial Time, and the space coordinate when i runs from 1 to 3. The inter-satellite time transfer link is used to measure the proper time difference with respect to a coordinate time between corresponding satellites. In addition to relativistic time transformation, the signal propagation delay caused by relativistic deflection of light and the satellite orbit correction introduced by relativistic celestial mechanics are also relativistic effects that need

to be considered in clock difference correction. The non-simultaneous emissions, unequal signal propagation delays, and unequal hardware delays between satellites jointly lead to the unequal observation values of two-way pseudo-ranges, local or inter-satellite time intervals between reception and emission. It is precisely this effect that is used in turn to construct various forms of clock difference observation equations. The schematic diagram of two-way time transfer between satellites is shown in Fig. 1. $\mathbf{X}$ is the satellite position vector, subscript A or B are satellite markers, t_1 and t_3 are respectively the coordinate times of satellites A and B when emitted, t_2 and t_4 are respectively the coordinate times of satellites A and B when received. T_{12} and T_{34} are respectively the signal propagation times from satellite A to B and from satellite B to A. Without considering the relativistic effect, and setting that the clock difference between satellite B and A at a coordinate time is $\Delta\tau$, when hardware delay, ionospheric delay, and other corrections are not considered temporarily, the clock difference equation based on the observations of the satellite local reception and emission times can be written approximately as

$$\Delta\tau \approx \frac{[\tau_r^B(t_2) + \tau_e^B(t_3) - \tau_e^A(t_1) - \tau_r^A(t_4)] + (T_{34} - T_{12})}{2}, \quad (1)$$

where τ is the proper time, whose subscripts e and r denote respectively emission and reception, and the superscripts A or B denotes satellites. The above equation shows that the observations can be regarded as the local times of emission and reception on two terminals, and also as the time intervals between reception and emission which perform respectively on two terminals $[\tau_r^B(t_2) - \tau_e^A(t_1)]$ and $[\tau_r^A(t_4) - \tau_e^B(t_3)]$. Especially, the observations can be regarded as $[\tau_r^B(t_2) - \tau_e^B(t_3)]$ and $[\tau_r^A(t_4) - \tau_e^A(t_1)]$, namely the local time intervals between reception and emission when two terminals adopt a convention to emit signals at the same local time $\tau_e^B(t_3) = \tau_e^A(t_1)$. The asymmetric propagation delay needs to be corrected regardless of observation methods. When the relativistic effects are considered and the two terminals emit signals at the same local time, the clock difference which is defined as the proper time difference between two satellites at emission coordinate time can be expressed as

$$\Delta\tau_c = \frac{1}{2}\{\Delta\tau^B - (\Delta\tau^A)^B + [\Delta\tau^A(t_4) + \Delta\tau^B(t_3) + T_{34} + \Delta\tau_d^A(t_4)]^B - [\Delta\tau^B(t_2) + \Delta\tau^A(t_1) + T_{12} + \Delta\tau_d^B(t_2)]^B\}, \quad (2)$$

where $\Delta\tau_c = \tau^B(t_1) - \tau^A(t_1)$, $\tau^N(t_n)$ (n runs from 1 to 4) is defined as the proper time on satellite N at coordinate time t_n , the observations $\Delta\tau^A$ and $\Delta\tau^B$ are respectively the local time intervals between reception and emission on satellite A and B, $\Delta\tau^N(t_n)$ is the correction of hardware delay at time t_n of emission or reception on satellite N, $\Delta\tau_d^N(t_n)$ is a correction of the delay owing to the effect of Doppler shift on the local time measurement of reception at time t_n , $(\Delta\tau^N)^M$ represents the time interval transformed from the local reference system of N to that of M, $(T_{34} - T_{12})^B$ is the asymmetric propagation time delay.

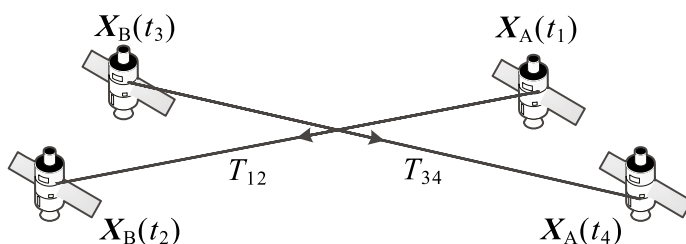


Fig. 1 Inter-satellite two-way time transfer

The time transformations needed to be considered in establishing a high-precision terrestrial coordinate time are already studied in great detail^[24–26], including the transformations among the ground local time, the satellite local time, terrestrial time (TT), and the Geocentric Coordinate Time (TCG). When establishing a high-precision terrestrial coordinate time and evaluating the performance of a space clock in autonomous long-term operation condition, the cumulative effects of time transformation need to be considered, and the precision needs to be improved as much as possible^[27]. Wolf et al.^[28] realize the time transformations among Earth’s surface local time, satellite local time, and TCG with an accuracy of 10^{-18} , of which the effects of Earth’s gravitational potential, second-order Doppler, external celestial bodies, and solid Earth tides are considered. Due to the fact that once time comparison takes up a few seconds at most, and that the correction of time interval observation can be completed by multiplying the corresponding coefficient of time transformation, time interval correction of 0.1 ps accuracy only requires a time transformation accuracy of 1×10^{-13} . According to the metric of geocentric celestial reference frame in the IAU 2000 resolutions, the contributions of the squared potential W^2 and the vector potential W^a (a runs from 1 to 3) is less than 10^{-18} ^[23] for the time transformations between the local times from the Earth’s surface to a high orbit satellite and TCG. Therefore, the time transformation for the clock difference correction needs mainly to consider the contributions of W and second-order Doppler to the metric. The time transformation between satellite local time and TT can be expressed as

$$\frac{d\tau}{dT} = \frac{d\tau}{dTCG} \frac{dTCG}{dT} \approx \left(1 - \frac{W}{c^2} - \frac{\mathbf{v}^2}{2c^2}\right) \left(\frac{1}{1 - L_g}\right), \quad (3)$$

where $\mathbf{v}$ is the satellite velocity, $1 - L_g$ is the coefficient of time transform between TT and TCG, $L_g = U_g/c^2$, and U_g is the equivalent gravitational potential of the geoid which includes the effect of Earth rotation. The inter-satellite time transformation can be calculated based on the above equation. The time transformation between the satellite local time and TCG and TT can reach the level of 10^{-17} ^[24] under the current condition of position and velocity determinations by applying a high-precision potential model including the Earth’s potential and second order Doppler effect, which supports the time transformation of time

interval observations at the level of fs for inter-satellite time transfer. It is sufficient high to meet the requirement for the clock difference correction at the level of order ps, therefore the time transformation in the relativistic framework is not key for the correction of clock difference, which will not go into much detail here. Besides the time transformation, as mentioned earlier, the clock difference correction needs to consider the signal propagation delay caused by the relativistic deflection of light and the satellite orbit correction owing to the relativistic celestial mechanics. Because the satellite orbit is offered by orbit determination, the relativistic effect still needed to be taken into account for the correction of clock difference is the signal propagation delay caused by the relativistic deflection of light.

2.2 The Correction of the Asymmetric Propagation Time Delay Caused by Satellite Motion

The correction of the asymmetric propagation delay needs to classify, derive, and reduce the combined expression based on the time delay expression of signal one-way propagation. If the relativistic effects are not considered, the time delay of signal propagation can be calculated using the method of the common follow-up problem. In the relativistic framework, the signal propagation trajectory is along the null geodesic equation with the proper time taking the extremum zero, and it is difficult to solve accurately when the high order terms of metric and gravitational potential are considered, which affects further the calculation of propagation delay. The Shapiro delay is the simplest delay of the relativistic light deflection, and it is derived in following several steps. The null geodesic equation is simplified under the conditions that the metric tensor is static and isotropic, the first order post-Newtonian approximation is used and the mass-monopole of gravitational potential is remained only. Then the motion integral and differential expression of time are obtained. Finally, the Shapiro delay can be derived by the integral of differential expression of time with the motion parameters^[28]. In this section the asymmetric propagation delay owing to satellite motion is derived, then the delay due to relativistic light deflection is analyzed and discussed in the next section. When the Shapiro delay is included, the signal one-way propagation time expressed with the parameters at the emission time in the geocentric non-rotating reference system is given by^[14]

$$T_{12} = \frac{D_{AB}(t_1)}{c} + \frac{\mathbf{D}_{AB}(t_1) \cdot \mathbf{v}_B(t_1)}{c^2} + \frac{D_{AB}(t_1)}{2c^3} \left\{ |\mathbf{v}_B(t_1)|^2 + \mathbf{D}_{AB}(t_1) \cdot \mathbf{a}_B(t_1) + \left[\frac{\mathbf{D}_{AB}(t_1) \cdot \mathbf{v}_B(t_1)}{D_{AB}(t_1)} \right]^2 \right\} + \frac{2GM_E}{c^3} \ln \frac{r_A(t_1) + r_B(t_1) + D_{AB}(t_1)}{r_A(t_1) + r_B(t_1) - D_{AB}(t_1)} + T_{AB}^{\text{og}} + T_{AB}^{\text{iono}} + T_{AB}^{\text{tropo}}, \quad (4)$$

where $\mathbf{D}_{AB}(t_n) = \mathbf{X}_B(t_n) - \mathbf{X}_A(t_n)$, is the relative position vector of satellite B with respect to satellite A at time t_n , $D_{AB}(t_n) = |\mathbf{D}_{AB}(t_n)|$, $\mathbf{v}_N(t_n)$ and $\mathbf{a}_N(t_n)$ are respectively the velocity and acceleration vectors of satellite N at time t_n , G and M_E are respectively universal gravitational constant and Earth's Mass. The second and third terms in the above equation

are respectively the second-order and third-order terms of c^{-1} caused by relative motion, the fourth term represents the Shapiro delay, T_{AB}^{iono} and T_{AB}^{tropo} represent respectively ionosphere and troposphere delays, T_{AB}^{og} represents the other delays of relativistic light deflection.

As mentioned earlier, the asymmetric propagation delay $(T_{34} - T_{12})^B$ is the difference between the inter-satellite two-way propagation times. If the asymmetric propagation time is calculated directly by substituting the expression of signal one-way propagation time, the combined expression will be complex because it contains various motion parameters of two satellites at respective emission times. Theoretical derivation shows that the c^{-1} and c^{-2} terms can be simplified by expressing the signal one-way propagation delays with the motion parameters at the same emission or reception time. Through reduction, parts of c^{-1} order terms are eliminated, the asymmetric propagation delay can be expressed by

$$\begin{aligned}
 T_{34} - T_{12} = & \frac{\mathbf{N}_{AB}(t_1) \cdot \mathbf{v}_{AB}(t_1) \Delta t}{c} + \frac{v_{AB}(t_1)^2 \Delta t^2}{2D_{AB}(t_1)c} + \frac{\mathbf{N}_{AB}(t_1) \cdot \mathbf{a}_{AB}(t_1) \Delta t^2}{2c} - \\
 & \frac{\mathbf{D}_{AB}(t_1) \cdot [\mathbf{v}_A(t_1) + \mathbf{v}_B(t_1)]}{c^2} - \frac{\mathbf{D}_{AB}(t_1) \cdot \mathbf{a}_A(t_1) \Delta t}{c^2} - \\
 & \frac{\mathbf{v}_{AB}(t_1) \cdot \mathbf{v}_A(t_1) \Delta t}{c^2} - \frac{\mathbf{D}_{AB}(t_1) \cdot \mathbf{j}_A(t_1) \Delta t^2}{2c^2} - \\
 & \frac{\mathbf{v}_{AB}(t_1) \cdot \mathbf{a}_A(t_1) \Delta t^2}{c^2} - \frac{\mathbf{v}_A(t_1) \cdot \mathbf{a}_{AB}(t_1) \Delta t^2}{2c^2} + \dots, \quad (5)
 \end{aligned}$$

where $\mathbf{N}_{AB}(t_1) = \mathbf{D}_{AB}(t_1)/|\mathbf{D}_{AB}(t_1)|$, $\Delta t = t_3 - t_1 = -\Delta\tau_c/k_B$, k_B represents the coefficient of time transformation from coordinate system to the local reference system of B. $\mathbf{v}_{AB}(t_1) = \mathbf{v}_B(t_1) - \mathbf{v}_A(t_1)$, $\mathbf{a}_{AB}(t_1) = \mathbf{a}_B(t_1) - \mathbf{a}_A(t_1)$, $\mathbf{j}_A(t_1)$ represents the derivative of acceleration of satellite A at time t_1 . The ellipsis represents the small terms which can be ignored. The above equation indicates that the correction of the asymmetric propagation delay needs to substitute the clock difference, therefore the calculation is an iterative. When the hardware delay is considered, the error of clock difference can decrease easily to the level of 10 ns after several steps iterative, and the correction error of the propagation delay owing to the clock difference error can decrease to 0.1 ps. The correction error of the asymmetry propagation delay is mainly caused by the errors of position and velocity determinations. The asymmetric propagation delay due to satellite motion is the maximum correction term of propagation delays, and its value, correction error, and the corresponding stability of correction residual error are evaluated by substituting the satellite simulation ephemeris. We write a computer program to simulate the ephemerides of low orbit satellite, medium orbit satellite, geosynchronous orbit satellite, and inclined geosynchronous orbit satellite (denote respectively by LEO, MEO, GEO, and IGSO). In the program, the non-rotating geocentric system is used, x axis points at vernal equinox, the x - y plane is the equatorial plane, and z axis points at the Earth's north pole. The Earth's central force is mainly considered, and the Binet formula for orbit integral is adopted for orbit calculation. The orbit altitudes of LEO, MEO, GEO, and IGSO are respectively set to be 400 km, 21528 km, 35786 km, and 35786 km, the eccentricities are all set to be 0.0005, the orbital inclinations

are respectively set to be $42^\circ, 55^\circ, 0^\circ$, and 55° , the right ascensions of ascending node are respectively set to be $109^\circ, 120^\circ, 110.5^\circ$, and 118° at initial time, the arguments of perigee and mean anomalies are set to be $\pi/2$ and $3\pi/2$, respectively. The trajectories of the above mentioned four satellites (LEO, MEO, GEO, and IGSO) and links among them at a time are shown in Fig. 2. The asymmetry propagation delays owing to relative motion are shown in Fig. 3, of which it is supposed that the accuracy of initial synchronization of inter-satellite two-way transfer reaches at the level of 10 ns which can be realized in a project easily. The discontinuity of curves shown in Fig. 3 illustrate the link interruptions owing to Earth occlusion. As shown in Fig. 3, the values of the asymmetric propagation delay owing to satellite motion are within $\pm 3 \mu\text{s}$.

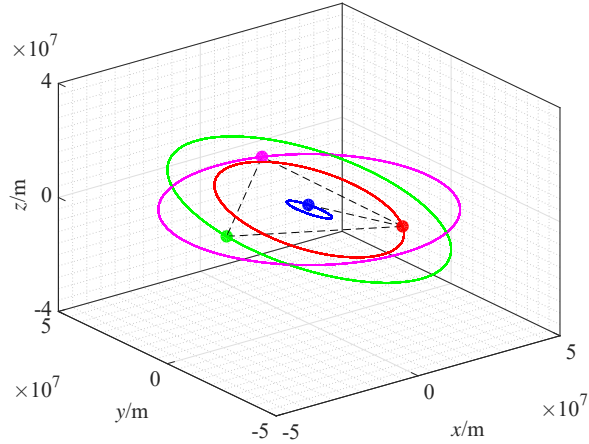


Fig. 2 Satellite orbit simulation (blue, red, purple, green dots and curves represent respectively LEO, MEO, GEO, IGSO satellites and trajectories)

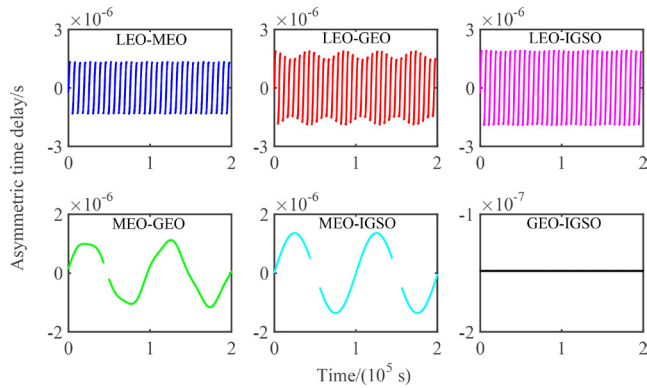


Fig. 3 Asymmetric propagation delay caused by satellite motion

The uncertainty on the correction of the asymmetric propagation time delay is evaluated quantitatively by simulating further the random and periodic noises of position and velocity determinations based on the simulation ephemeris and substituting them to calculate. Assuming that the noises of position and velocity are random, the variances of the orbit errors in the radial, tangential, and normal directions are all set to be 1 m, and the variances of the radial, tangential, and normal components of velocity error are all set to be $0.0001 \text{ m} \cdot \text{s}^{-1}$. The difference sequence between the asymmetric propagation time calculated by substituting the simulation orbit and velocity sequences and one calculated by substituting the orbit and velocity sequences with additional simulation errors represents the residual error sequence, as shown in Fig. 4. It shows that the fluctuations of the residual error sequence of propagation delay correction are less than 1 ps under the condition of orbit determination and velocity measurement mentioned before. The amplitude of the residual error of propagation delay correction is independent of noise type of orbit, therefore the result under random noise is also suitable for the one under periodic noise, however the additional stability of the residual error of time delay correction is affected by the noise type of orbit determination and velocity measurement. The time Allan deviation (the square root of time Allan variance, TDEV) of the residual error sequence of propagation time correction is namely the corresponding additional stability, which represents quantitatively the effect of propagation time correction on the time transfer link. Assuming that the noise types of orbit determination and velocity measurement are random and periodic, the TDEV of the residual error sequence of propagation time correction with respect to the time transfer between GEO and IGSO is shown in Fig. 5. In the calculation it is assumed that there are three periodic terms of orbit errors, where the periods are respectively 1, 0.5, 0.33 times the orbital period, the corresponding amplitudes of orbit errors are respectively 0.5 m, 0.4 m, and 0.3 m, and the corresponding amplitudes of velocity errors are respectively $0.00005 \text{ m} \cdot \text{s}^{-1}$, $0.00004 \text{ m} \cdot \text{s}^{-1}$, and $0.00003 \text{ m} \cdot \text{s}^{-1}$. It shows that when the noises of orbit determination and velocity measurement are random, the TDEV $\sigma_x(\tau)$ (the subscript x refers in particular to the observations of normalized time or phase) of the residual error sequence of the propagation time correction is better than 0.1 ps/s and proportional to $\tau^{-1/2}$, namely $\sigma_x(\tau) \propto \tau^{-1/2}$, which indicates that the noise type is white phase noise and the corresponding stability decreases faster than the ones of white frequency, flicker walk frequency, and random run frequency noise with increasing averaging time. When the averaging time is in the range 1–1000 s, the $\sigma_x(\tau)$ caused by the periodic noises of orbit and velocity is smaller than the ones caused by the random noise, and it is slightly larger at the averaging time 10000s–100000 s, however the maximum is smaller than 10^{-14} s.

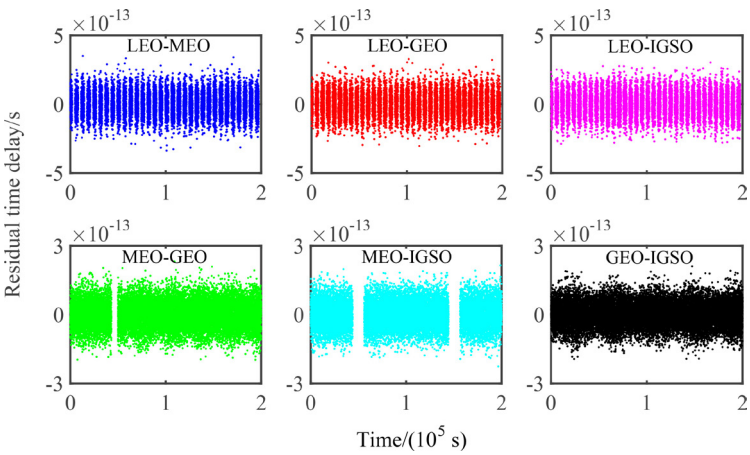


Fig. 4 The correction residuals of asymmetric propagation delay

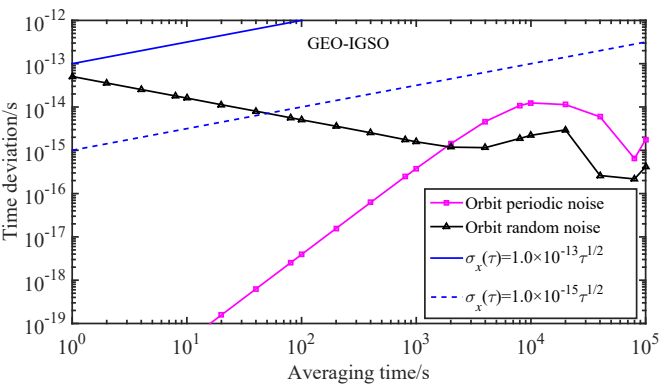


Fig. 5 The TDEV of the correction errors of asymmetric propagation delay

In addition to the quantitative analysis of the accuracy of the propagation delay correction, the link design of inter-satellite time and frequency transfer also needs to deduce conversely the requirements of position and velocity determinations according to its accuracy requirements. Because the asymmetric propagation delay caused by satellite motion is most sensitive to the changes of satellite orbit and velocity, it is most suitable to be used to deduce the requirements on satellite position and velocity determinations. The phase noise of atomic clock is mostly the white frequency noise, namely TDEV is proportional to $\tau^{1/2}$. Considering the type of the correction residual error of time delay, that is, the trend of $\sigma_x(\tau)$ curve with respect to the averaging time τ , when moving $\sigma_x(\tau)$ to make it less than the TDEV of the atomic clock at any averaging time, we can determine the closest point $\sigma_x(\tau_c)$ between it and the stability curve of the atomic clock. As shown in Fig. 5, when the TDEV of atomic clock is $\sigma_x(\tau) = 1.0 \times 10^{-13} \tau^{1/2}$ s, the TDEV closest points between the atomic

clock and the propagation delay correction residual under the conditions of orbital white noise and periodic noise are respectively $\sigma_x(1\text{ s})$ and $\sigma_x(10000\text{ s})$. Then, according to the relationship between $\sigma_x(\tau)$ and the residual amplitude of propagation time delay correction, the maximum allowable values of the residual amplitude and orbit determination are evaluated in turn. If the stability curve of the propagation delay correction residual is only smaller than the one of the atomic clock for medium to long averaging times, the closest point of the two curves is the intersection, which reduces the demands for position and velocity determinations. As an example, when the orbit noise is random, there is a relationship between the propagation delay correction error and its stability $\delta T_c \leq 4\sigma_x(1\text{ s})$, as shown in Fig. 4 and Fig. 5. And while setting that $\sigma_x(1\text{ s}) < k_1\sigma_a(1\text{ s})$, therefore $\delta T_c \leq 4k_1\sigma_a(1\text{ s})$, where $\sigma_a(1\text{ s})$ is the time stability of the atomic clock at time 1 s, and $0 < k_1 < 1$ is the proportional coefficient which is set from the performance budget of the time and frequency transfer link, and by considering comprehensively the conditions of hardware delay correction and time measurement error. The link design of inter-satellite time and frequency transfer can deduce conversely the requirements of position and velocity determinations according to the above method by considering the noise type of position and velocity determinations. For instance, Table 1 displays the requirements on orbit determination and velocity measurements if the asymmetric propagation delay correction is less than 0.1 ps, δR , δT , and δN are respectively the allowed errors of radial, tangential, and normal components of orbit determination, δv_R , δv_T , and δv_N are respectively the allowed errors of radial, tangential, and normal components of velocity measurement, the values before and after the short horizontal line represent respectively the requirements of the two satellites which perform time comparison. As shown in Table 1, if the residual time error of correction reach the level of 0.1 ps, the requirements on the radial, tangential, and normal components of orbit determination are respectively 0.7–6 m, 0.5–1 m, and 1–60 m, and the requirements on the radial, tangential, and normal components of velocity measurement are respectively $0.07\text{--}0.3\text{ mm} \cdot \text{s}^{-1}$, $0.1\text{--}0.8\text{ mm} \cdot \text{s}^{-1}$, and $0.1\text{--}7\text{ mm} \cdot \text{s}^{-1}$.

Table 1 The requirement evaluations of orbit determination and velocity measurement for inter-satellite two-way time transfer (correction residual of propagation time delay: 0.1 ps)

Inter-satellite link	$\delta R/\text{m}$	$\delta T/\text{m}$	$\delta N/\text{m}$	$\delta v_R/(\text{mm} \cdot \text{s}^{-1})$	$\delta v_T/(\text{mm} \cdot \text{s}^{-1})$	$\delta v_N/(\text{mm} \cdot \text{s}^{-1})$
LEO-MEO	1–0.7	0.5–0.5	5–3	0.3–0.2	0.2–0.7	0.8–3
LEO-GEO	2–0.7	0.5–0.5	3–1	0.2–0.1	0.2–0.8	0.2–1
LEO-IGSO	2–0.7	0.5–0.5	7–3	0.2–0.1	0.2–0.7	0.5–3
MEO-GEO	2–1	0.7–1	2–2	0.07–0.07	0.1–0.2	0.2–0.2
MEO-IGSO	2–1	0.7–1	60–50	0.07–0.07	0.1–0.2	3–7
GEO-IGSO	6–6	0.7–0.8	2–2	0.3–0.3	0.4–0.4	0.2–0.1

2.3 The Correction of the Time Delay Caused by the Relativistic Deflection of Light

In the relativistic framework, the propagation trajectory of electromagnetic signal is the solution of the null geodesic equation. Supposing that the Earth's gravitational potential is rotational symmetry, ignoring the Earth's rotation effect, using the first-order post-Newtonian approximation, and remaining the Earth's mass-monopole of gravitational potential only, the trajectory of electromagnetic signal in the vicinity of Earth obey the hyperbola^[28]. However, it is difficult to obtain the motion integral or solve the null geodesic equation for calculating the light propagation trajectory and propagation delay when Earth's multipole moments and rotation effect are considered. The time transfer function can be used to calculate the time delay caused by various kinds of gravitational potentials to order c^{-4} ^[29–31] without accurately solving the null geodesic equation. The metric takes the first-order post-Newton approximation, $G_{\alpha\beta} = \eta_{\alpha\beta} + h_{\alpha\beta}$, $\eta_{\alpha\beta}$ is the Minkowski metric, $h_{\alpha\beta}$ is the perturbation term of the metric, α, β run from 0 to 3. According to the condition that the proper time is 0, that is, $ds^2 = c^2 d\tau^2 = -G_{\alpha\beta} dX^\alpha dX^\beta = 0$, it can be obtained by substituting the metric and approximating to the first-order Taylor expansion

$$dt \approx \frac{|d\mathbf{X}|}{c} \left[1 + \frac{1}{2} (h_{00} + 2 h_{0i} n^i + h_{ij} n^i n^j) \right], \quad (6)$$

where j runs from 1 to 3, $n^i = v^i/c$, Einstein's summation convention is implied, the time and space coordinates are separated, and the propagation time can be calculated by integrating along the trajectory. The time transfer function is defined as the integral of $G_{\alpha\beta}(dX^\alpha/d\lambda)(dX^\beta/d\lambda)$ along the trajectory of signal propagation^[29], where λ is the affine parameter. Because the proper time takes the extremum zero, the integral is equal to zero. It constructs a variable problem with fixed endpoints ($J[y] = \int_{x_0}^{x_1} f(x, y, y') dx$), and the Euler-Lagrange equation ($f_y - d(f_{y'})/dx = 0$) can be used. Linet et al.^[29] derive an another expression of geodesic equation of light propagation, and prove that if the integral trajectory is expressed as a straight line plus a disturbance, the disturbance itself is on the order of c^{-2} , and its effects on integrand $G_{\alpha\beta}(dX^\alpha/d\lambda)(dX^\beta/d\lambda)$ and the propagation time delay are respectively order c^{-4} and c^{-5} , which can be ignored when calculating the propagation time delay to order c^{-4} . Integrating Eq. (6) along the straight line part of signal propagation in Euclidean space, the time transfer function^[29] with c^{-4} accuracy is given by

$$\begin{aligned} T_s(t_A, \mathbf{X}_A, \mathbf{X}_B) \\ = \frac{R_{AB}}{c} + \frac{R_{AB}}{2c} \int_0^1 [h_{00}(X'(\lambda)) + h_{ij}(X'(\lambda)) N^i N^j + 2h_{0i}(X'(\lambda)) N^i] d\lambda, \end{aligned} \quad (7)$$

where $X'(\lambda)$ represents the straight line between terminals A and B, $X^{i'}(\lambda)$ is the component of $X'(\lambda)$, $N^i = [X_B^{i'}(\lambda) - X_A^{i'}(\lambda)]/R_{AB}$, R_{AB} is the distance of the straight line between A and B, $X^{i'}(\lambda) = X_A^i + (X_B^i - X_A^i)\lambda$, $0 \leq \lambda \leq 1$. Soffel et al.^[30] also derive the propagation time based on the time transfer function. Overall, the method of time transfer function

expresses the propagation time delay caused by the relativistic deflection of light as the integral of $h_{\alpha\beta}$, and because $h_{\alpha\beta}$ can be expressed as the polynomial of c^{-1} , therefore kinds of delays caused by various gravitational potential can be calculated separately. The time delays of relativistic deflection of light are evaluated by using the expression derived by Linet et al.^[29] and substituting the ephemeris. Fig. 6 shows the relativistic time delays of one-way link vary with satellite motion, including the delays caused by the mass-monopole, quadruple moment, and Earth's rotation. Fig. 7 shows the asymmetry relativistic delays, whose maximum value is at the level of 10 fs, which meets the demand of correction of the asymmetry propagation time delay at the level of 0.1 ps.

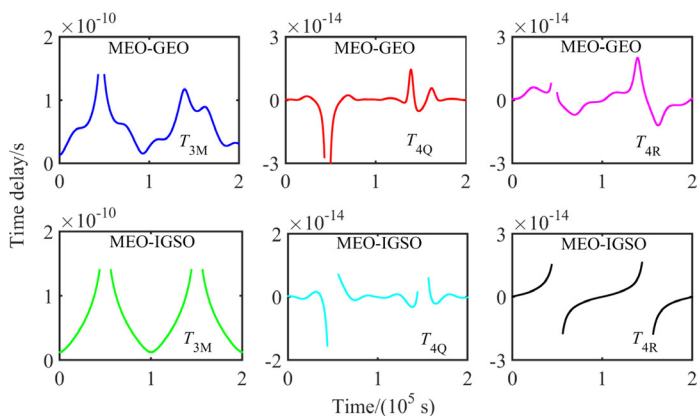


Fig. 6 Propagation delays caused by relativistic deflection of light (T_{3M} , T_{4Q} , T_{4R} represent separately the time delays caused by Earth's mass, quadruple moment, and spin potential)

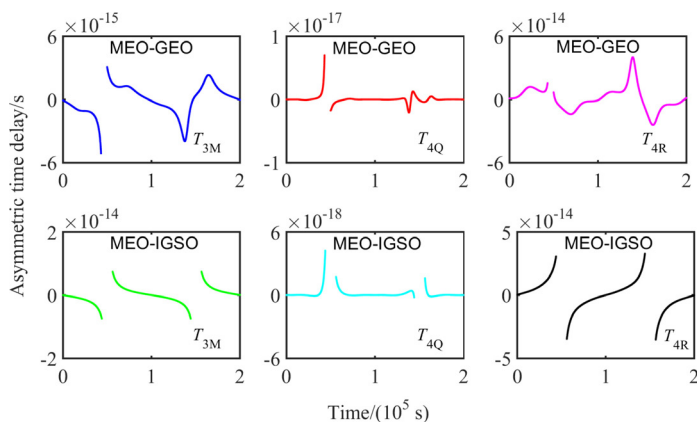


Fig. 7 Asymmetric propagation delays caused by relativistic deflection of light

In addition to the above mentioned time delays caused by the relativistic deflection of light, other time delays related to the potentials of solid Earth tides, Sun mass, and Moon mass etc can be also evaluated by the time transfer function. Han et al.^[31] evaluated the relativistic time delays of signal propagation caused by ocean tides, inertial term etc for the satellite-ground link, and the values are at the level of 10^{-16} s or less. This method can be extended to the calculation of the relativistic time delay of inter-satellite time transfer, and it infers that these relativistic time delays have no effect on the time transfer of order 0.1 ps.

On the whole, the asymmetry propagation time delay is caused by the asymmetry propagation trajectory of inter-satellite, and is one of the primary terms for clock difference correction. The relative motion is the primary reason that causes the asymmetry propagation time delay, and it combines with relativistic deflection of light and non-simultaneous emission together to lead to other asymmetry propagation delays which need to be corrected separately. We establish a correction method of the asymmetric propagation delay, including reducing the asymmetric propagation delay caused by satellite motion, calculating respectively the light deflection delays caused by the potentials of Earth's mass monopole potential, quadrupole moment and spin, and evaluating the values, residuals, and additional stabilities of the asymmetric propagation delay correction. It indicates that under the current precise orbit determination conditions, the asymmetric propagation delay can be corrected to the level of 0.1 ps.

3. THE EFFECT OF DISCONTINUOUS MEASUREMENT OF THE INTER-SATELLITE TIME TRANSFER ON ALLAN DEVIATION AND THE CORRESPONDING CORRECTION METHOD

Atomic clocks on different satellites obtain the clock and frequency difference sequences between each other through inter-satellite time and frequency transfer links. Due to the fact that the variances of clock or frequency difference sequence are difficult to accurately characterize the drift of frequency sources, their calculated values may diverge with increasing statistical data, making it difficult to accurately predict the cumulative time error of clock or frequency sources after autonomous operation for a period of time. The theory of time and frequency statistics predicts the time keeping deviation of atomic clocks by calculating the Allan variance of clock or frequency difference sequences. The Allan variance for frequency data is defined as

$$\sigma_y^2(\tau) = \frac{1}{M-1} \sum_{p=1}^{M-1} \frac{(\bar{y}_{p+1} - \bar{y}_p)^2}{2}, \quad (8)$$

where the subscript y of $\sigma_y(\tau)$ refers in particular to the observation of relative frequency, the frequency difference sequence is divided into M sampling intervals according to the averaging time τ , $\bar{y}_p$ is the average value of fractional frequency difference over the p th sampling interval. Allan variance can be also expressed as an integral of frequency auto-

correlation function or power spectral density, which facilitates the analysis of frequency stability performance of frequency sources from the perspective of power law noise.

As mentioned earlier, the stability (Allan variance) calculation is almost unaffected when the averaging time τ is much less than the measurement period T in the case of a measurement sequence with periodic missing data; Barnes et al.^[20, 21] and Wang et al.^[22] adopted the coefficient $B_2(T, \tau)$ to correct the stability calculation result when the averaging time τ is on the same order with the measurement period T , and is less than T in the case of a measurement sequence with periodic missing data. For another case of τ larger than T or a measurement sequence with irregular missing data, the average value of relative frequency difference $\bar{y}_p$ in Eq. (8) cannot be accurately calculated due to the missing sampling points, making it difficult to accurately calculate the Allan variance. Theoretical calculation shows that the impact of missing data on Allan variance can be attributed to the changes of the statistical weights of the autocorrelation function by expanding the autocorrelation function expression of Allan variance. Combined with the analytical form of autocorrelation function of the power law noise, the deviation of Allan variances for different types of noise can be calculated separately. Under the condition of discrete sampling, the formula of Allan variance for frequency data can be written as

$$\sigma_y^2(\tau) = \frac{1}{m^2} \sum_{m_1=1}^m \sum_{m_2=1}^m R_y[(m_1 - m_2)\tau_0] - \frac{1}{m^2} \sum_{m_1=1}^m \sum_{m_2=m+1}^{2m} R_y[(m_1 - m_2)\tau_0], \quad (9)$$

where R_y represents the autocorrelation function of relative frequency difference, and the subscript y refers in particular to the observation of relative frequency, $m = \tau/\tau_0$, τ_0 is the sampling time interval. By setting $m_1 - m_2 = k$ and $q = |k| + 1$, the formula of Allan variance can be rewritten as

$$\sigma_y^2(\tau) = \frac{1}{M-1} \sum_{p=1}^{M-1} \sum_{q=1}^{2m} \frac{N_{pq}}{m^2} R_y[(q-1)\tau_0], \quad (10)$$

where N_{pq}/m^2 represents the normalized probability of $R_y[(q-1)\tau_0]$ in the term $(\bar{y}_{p+1} - \bar{y}_p)^2/2$ expressed in Eq. (8). The above equation indicates that Allan variance is calculated by sequentially calculating the probability of the autocorrelation function $R_y[(q-1)\tau_0]$ in the terms $(\bar{y}_{p+1} - \bar{y}_p)^2/2$ at each sampling interval, then averaging the $M-1$ values, and finally, it is expressed as a weighted sum of $R_y[(q-1)\tau_0]$.

The autocorrelation function is the Fourier transform of power spectral density, and in the statistical theory for the time and frequency filed, the power spectral density of power law noise is often expressed as $S_y(f) = h_\gamma f^\gamma$, where f represents the noise frequency, $\gamma = 2, 1, 0, -1$, and -2 respectively represent white phase, flicker phase, white frequency, flicker frequency, and random walk frequency noise. When γ is not an integer, by applying the integral formula^[32] $\int_0^\infty x^{-z} \cos bx dx = (\pi/2)[b^{z-1}/\Gamma(z) \cos(\pi z/2)]$ which was derived from Γ function, one can obtain $R_y(\tau_1) \propto |\tau_1|^{-\gamma-1}$ ^[33]. When γ is equal to 0, positive, and negative integer, the calculation of $R_y(\tau_1)$ is difficult, and the above power law relationship between

$R_y(\tau_1)$ and τ_1 is also used approximately. Due to the limited bandwidth, $R_y(0) = R_y(1/w_B)$, where w_B is the sampling instrument bandwidth^[33].

Under the condition of missing data, if the sampling points are still divided by the averaging time τ , the missing data points are directly discarded, and the average value of the relative frequency difference in each effective sampling interval and the frequency stability are calculated successively, it will cause the calculation error of stability, because it is equivalent to making the weight of $R_y(\tau_1)$ of frequency stability (τ_1 takes values $0, \tau_0, 2\tau_0, \dots, (2m-1)\tau_0$ in the case of discrete sampling) changed, which no longer meets the definition of Allan variance. Fig. 8 and Fig. 9 show the impacts of periodic data loss on stability calculation. The period of the simulation observation data is the period (5714 s) of Earth orbital revolution of low Earth orbit (altitude of about 400 km), and the original data are from the test data of three active hydrogen clocks (denoted respectively by HM1, HM2, and HM3). Fig. 8 shows the frequency Allan deviations in the cases of missing data at 95% (denoted by DCC) and not missing (denoted by CC). Fig. 9 shows the relationship between frequency Allan deviations of HM3 and the proportions of periodic effective data, in which the percentages in the legend which is in the upper right corner represent the proportions of effective data. It can be seen from the above two figures that under the condition of high proportion of missing comparison data, the medium-term stability deviation can be up to three orders of magnitude, and the main noise type is white frequency noise, while the long-term stability deviation is less, and the main noise types are flicker and random walk frequency noise.

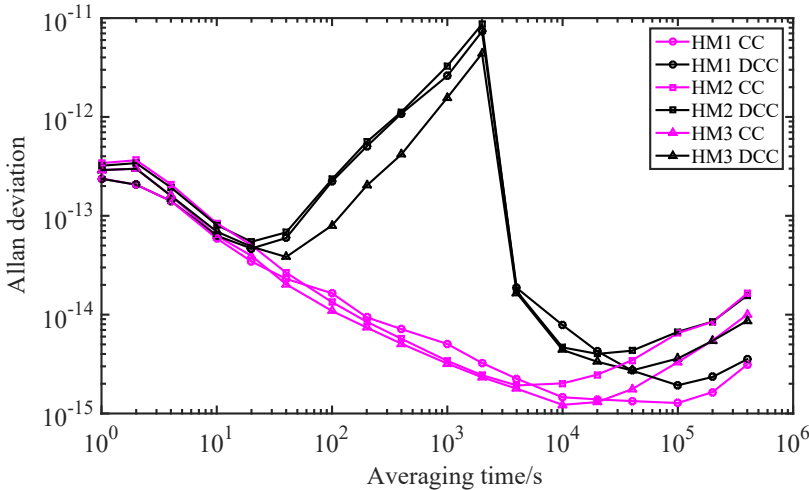


Fig. 8 The frequency Allan deviation under 95% missing and no missing data conditions

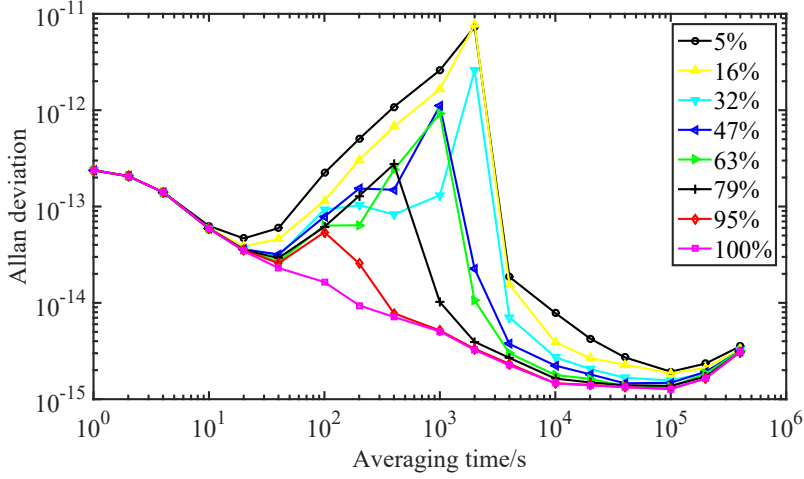


Fig. 9 The relationship between frequency Allan deviation and the proportion of periodic effective data for HM3

The type of noise and the proportion of missing data are the two dominant factors affecting stability calculation. Atomic clocks have different types of noise or present a combination of various types of noise at different averaging times. The noise types or their weight distributions of onboard atomic clock at each averaging time can be obtained based on ground test results. By calculating respectively the correction coefficients for the stability of the data presenting a power law noise under missing data condition, they can be used to correct the stability of the measurement data at different averaging times. Based on the autocorrelation function representation of Allan variance mentioned above, a scale factor is defined as

$$k_2 = \sqrt{\frac{\sum_{\tau_1=0}^{2m-1} P_R(\tau_1) R_y(\tau_1)}{\sum_{\tau_1=0}^{2m-1} P_0(\tau_1) R_y(\tau_1)}}, \quad (11)$$

where $P_R(\tau_1)$ and $P_0(\tau_1)$ are respectively the normalized probabilities of $R_y(\tau_1)$ under the conditions of missing and non-missing data, and are calculated by programming. k_2 represents the stability deviation of the data presenting a power law noise under the condition of missing data. As an example, Fig. 10 shows the $R_y(\tau_1)$ probability with respect to the Allan deviation for frequency data at 10000 s under the condition of different proportions of effective data, where the percentages in the legend which is in the upper right corner represent the proportions of effective data, and the data period is 5714 s. According to Eqs. (9) and (10), the probability of $R_y(\tau_1)$ may take a negative value, but it will not affect the calculation. It can be found from Fig. 10 that when the missing data probability is high, the probability of $R_y(\tau_1)$ has extreme values at different autocorrelation times τ_1 . When $\gamma = 2, 1, 0, -0.5, -1.5$, and $\tau = 10000$ s, the examples of k_2 are shown in Table 2, where the

percentages represent the proportions of effective data. The simulation calculation uses the periodic missing data, but it can be extended to the case with non-periodic missing data. In the above calculation, when γ is negative, -0.5 and -1.5 are used while -1 and -2 not, this is because Γ function is infinite when γ is a negative integer, which make it difficult to calculate the analytical form of autocorrelation function.

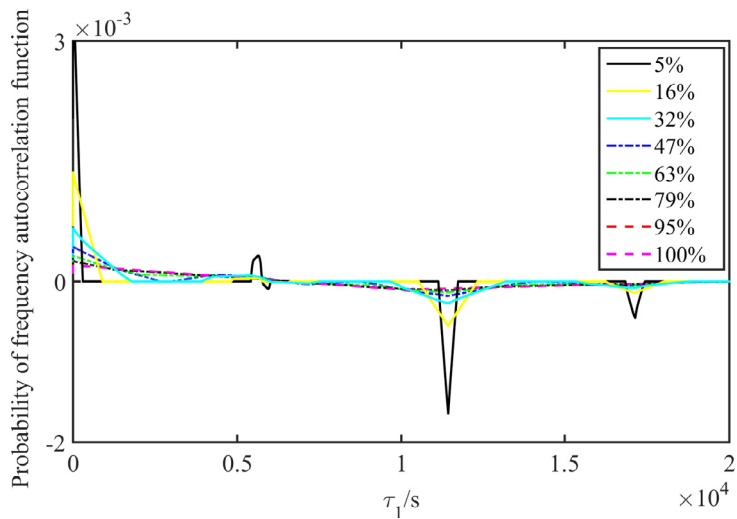


Fig. 10 The relationship between the probability of frequency autocorrelation function and the proportion of periodic effective data

Table 2 The correction coefficient k_2 value of frequency Allan deviation with missing data ($\tau = 10000$ s)

k_2 γ	R^a				
		5%	32%	63%	95%
0, 1, 2		4.52	1.83	1.27	1.04
-0.5		4.29	1.78	1.25	1.03
-1.5		1.17	1.08	1.02	1.00

^a The proportion of the effective frequency data.

As can be seen from the above table, when $\gamma = 2, 1, 0$ and the effective data accounts for 5%, 32%, 63%, and 95%, the calculated values of 10000 s Allan deviation increase by 3.52, 0.83, 0.27, and 0.04 times, respectively. Fig. 8 shows that when the effective data proportion is 5%, the average value of 10000 s stability of HM1 and HM3 increases by 3.4 times, which is approximately consistent with the result 3.52 in Table 2, of which because HM2 presents the combination of multiple noises at 10000 s, and its increasing multiples of 10000 s stability are not used for analysis. As can be seen from Fig. 9, when the proportions of effective data are 5%, 32%, 63%, and 95%, the corresponding increasing multiples of 10000 s stability of HM3 are respectively 4.13, 0.87, 0.20, and 0.07, which are also approximately consistent with the results in Table 2. Table 2 also shows that under the condition of missing data, the deviations of stability are larger when $-\gamma - 1 > 0$ ($\gamma < -1$) and smaller when $-\gamma - 1 < 0$ ($\gamma > -1$), this trend is also consistent with the calculation results in Figs.8 and 9. Figs. 8 and 9 also show that for the active hydrogen atomic clock, the stabilities at the averaging time of 100–10000 s are dominated by the white frequency noise ($\gamma = 0$), and the deviations of the calculated values under the condition of missing data are large; the stability at the averaging time of one day is dominated by flicker frequency ($\gamma = -1$) and random walk frequency noise ($\gamma = -2$), and the deviation of calculated value under the condition of missing data is small. Therefore, the data missing with high proportion mainly affects the stability calculation of white phase noise, flicker phase noise, and white frequency noise, while it has little effect on the stability calculation of flicker frequency noise and random walk frequency noise. Because the mid-term stability of atomic clock is mostly dominated by white frequency noise, the data missing with high proportion seriously affects the stability calculation. In order to decrease the deviation of Allan variance, ground or satellite relays should be used to reduce the proportion of data missing, or stability correction should be performed. In conclusion, stability correction can skip missing frequency or time difference data points and directly calculate the average value over each effective sampling interval, further calculate the initial value of Allan deviation; then calculate the correction coefficient k_2 based on condition of the missing data; finally, multiply the initial value of Allan deviation by k_2 and obtain the corrected result.

4. CONCLUSIONS

Besides hardware delay, the asymmetric propagation delay and missing data are the two main factors affecting clock difference correction of inter-satellite time transfer. For the correction of the asymmetric propagation delay, the measurement method of clock difference is established and analyzed firstly; then the asymmetric propagation delay caused by the relative motion of satellites is derived and reduced, the delays caused by relativistic deflection of light are also calculated respectively by using the time transfer function method, and the stability of residual error of the propagation time delay correction is evaluated; finally, the constraints on satellite position and velocity determinations are inferred according to

the link accuracy. In conclusion, a comprehensive method of the propagation delay correction and evaluation for inter-satellite time transfer is proposed. It indicates that under the current condition of orbit determination (satellite orbit position and velocity accuracy are respectively 1 m and $0.1 \text{ mm} \cdot \text{s}^{-1}$), the correction accuracy of inter-satellite asymmetric propagation delay can reach the order of 0.1 ps. For the stability correction with missing data for inter-satellite time transfer, the expression of Allan variance based on the autocorrelation function is extended, and the impact of missing data on the stability of different types of noises is discussed. It illuminates that the high proportion of missing data has a greater impact on the stability calculation of white phase noise, flicker phase noise, and white frequency noise, while the impact on the stability calculation of flicker frequency noise and random walk frequency noise is smaller. Finally, a method for calculating the stability deviation of time and frequency transfer is proposed from the perspective of weighted statistics of autocorrelation function, and the Allan deviation correction coefficient k_2 is introduced, which can be used to correct the stability of inter-satellite time and frequency transfer.

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